Personal Space Weather Station

Descriptions & Definitions

Version 0.10

2020-07-01 UTC

Time Zone Definition
Node Definition
Data Base Contents
Saved Filename Structure
System Directory and File Structure
Metadata Contents
Saved File Examples
A Day in the Life of a Grape Gen 1

Table of Contents

Can’t see to get this to work yet…. argh

Edit History

- **Rev 0.00** 2020-04-28 by JCGibbons / N8OBJ — Initial release
- **Rev 0.01** 2020-05-01 by JCGibbons / N8OBJ — mods per Nathaniel
- **Rev 0.02** 2020-05-05 by JCGibbons / N8OBJ — added content about filename structure , nodes
- **Rev 0.03** 2020-05-05 by JCGibbons / N8OBJ — mods to directory structure, node info
- **Rev 0.04** 2020-05-06 by JCGibbons / N8OBJ — increased verbiage WRT project assumptions
  removed [minus] from illegal chars in filename spec
  modified top descriptive name of file
  added metadata definition / file format info
- **Rev 0.05** 2020-05-22 by JCGibbons / N8OBJ — refined time / timestamp definition
  refined node definition
  file name adjustments
  directory path adjustments
- **Rev 0.06** 2020-05-26 by JCGibbons / N8OBJ — Added PSWS file structure naming
- **Rev 0.07** 2020-06-04 by JCGibbons / N8OBJ — Started work on official formatting – more info in filesystem

- **Rev 0.08** 2020-06-04 by Bill Engelke / AB4EJ — Reviewed and added useful comments
- **Rev 0.09** 2020-06-18 by JCGibbons / N8OBJ — More file structure updates / Grape Gen 1 operation narrative
- **Rev 0.10** 2020-07-01 by JCGibbons / N8OBJ — Formatting work - TOC

Purpose of Document

This document is intended to define naming conventions, file structures and descriptions for the Personal Space Weather Station (PSWS). This includes the file naming structure, Node numbers, file structures, conventions, Metadata formats and header formatting for both the Low Cost (Grape) Personal Space Weather Station as well as the High Performance (Tangerine) Personal Space Weather Station. Inclusive in this definition would be personal ham stations using amateur, commercial, custom or any combination thereof of data collection equipment to support the PSWS mission.

It was requested that the file naming be informative of its contents by date, Node number, Grid Locator, and other info such that rudimentary sorting of the data files (and the information contained within) could be quickly assessed by the simple inspection of the filename itself.

This proposal is an effort to define the file and naming structure that will be used for the PSWS project.

It is assumed that the person looking at these files is knowledgable about this PSWS project and its intended purpose and thus understands what, why and how the data was collected.

Note:
This document is intended to be a work in progress as only the entire group knows all aspects of this project and what the needs are for each aspect of the project.

Time Standard / Definition

ALL PSWS data collection will be done on a one day Coordinated Universal Time (UTC) or Zulu (Z) day cycle. This will be done so as to avoid confusion with time zones.

All collected data will contain (where possible) the UTC Date & Time using the ISO 8601 format (again, if possible) preferrably in the filename itself, but again dependent on the type of data being collected.

In the Low Cost (Grape) PSWS, the ISO UTC date will also be included in the first line of the file for data processing purposes (along with other metadata for assisting in daily data plotting). If data collection is interrupted, the remaining data for that UTC day and type will simply be appended to the already existing data file. For the PSWS data submissions, each filename will have an ISO formatted UTC time stamp associated with each entry along with the Node number and other station related info.

In the High Performance Tangerine PSWS, the file naming and time stamping of collected data is done according to the DRF or RFS file definition. The Tangeringe team will determine the filename formatting to be used for data collection as well as the timestamp converntions used.

For general data acollection files for either system, it is preferred to have a date and time in the filename for easy collection time identification. The specifics of how this will be accomplished will be left up to the developer and reviewed by the team.

Node Definition / Format

A Node Number is defined to be a number (N followed by 5 digits) assigned to a unique computing unit at any given location that is responsible for the collection, control and submission of data to support the PSWS experiment. This computer will be capable of receiving commands from the database directly (when it contacts the database for heartbeat / query for command) and executing them accordingly.

FORMAT: N12345

The Node Number (N followed by 5 numeric digits) will:

1. Define the unique computer at the location contained registered for the node number assigned
2. Be assigned and controlled by the database control system
3. The data set presented from this node number must be able to stand on its own to represent the stations experiment's data submission [for subsequent analysis] to support the PSWS project. The data set submitted for this node number is considered to be unique to this node and not duplicated by another node (i.e. a given data set [or sub-set] collected by node N00001 is not to be reported by N00002). Exceptions to this will be experiment dependent and
4. Be the fundamental data base reference index for all metadata related to this computers's data collection activities

For the majority of the stations, the node number assignment will be straightforward. One node number is assigned for a given data collection 'computer'. A 'computer' is defined as the central computing resource that is running/controlling the data collection tasks assigned to it (doppler shift measurements, Magnetometer, Temperature, Lightning, RF data collection, etc) and is also responsible for the data exchange to the central database. Multiple data collection computers (hence multiple nodes) can exist at any given physical collection site, but all computers running a unique experiment (computer) will be assigned a separate node number. This node per computer is done so commands can be sent to any collection node to modify its operation (when the node queries the central collection as a heartbeat and a query for a command file to upload). The Metadata included in the data collection files will refine the information about the experiment (if needed).

If. for a given node number. multiple same date station data sets (for a given UTC day) are submitted to the database, the last submission will be used and the previous data set will be kept under version control. This will give a mechanism to allow for correcting data collection errors (if they arise).

Initial Assignment of Node Numbers (and what they mean)

- N00000 – Test Node - ALL data contained is for testing any PSWS - not to be used in real DB
- N00001 – N00099 – Reserved for initial development/testing teams for all PSWS's. Data is valid
- N00100 – N99998 – Normal "citizen" user installed nodes for any PSWS installation
- N99999 – Special node assignment for a universal broadcast command structure to all nodes

Special significance will be given to N00000 as a test node for coding, testing and development by the development teams. It is not assigned to ANY working station.

Special significance will also be given to N99999 as it will be iused as a broadcast to all nodes for commands sent from the central data control center to any station node.

The database should be able to sort and view all aspects of these database fields from a database engine designed to do these inquiries. Security to see contact info needs to be controlled based on who is using the query engine.

Node DataBase Contents Definition

The data for each collection node contains (but is not limited to) the following information:

1. Node number (N00001 – N99998)
2. City and State
3. Latitude, Longitude, Elevation
4. GridSquare (Maiden-Head Locator) AA00aa (full 6 digits) – [auto generated from Lat / Long]
5. Radio(s) Make and Model or custom description
6. Radio ID used in file naming for submitted data to central server
7. Antenna Description (Make and Model if commercial, description otherwise)
8. Magnetometer information (Make and Model if commercial, description otherwise)
9. Temperature Sensor(s) information (Make and Model if commercial, description otherwise)
10. Lightning Sensor information (Make and Model if commercial, description otherwise)
11. Station information – System type and software version(s)
12. tbd Sensor information (future additions)

Data that needs a higher security level (Contact Info) to view but is included WRT the node:

13. Contact Callsign
14. Contact Name
15. Contact full Address
16. Contact Cell Phone
17. Other info TBD

PSWS Filename Structure

Examples for the Grape and Tangerine PSWS

YYYY-MM-DDTHHMMSSZ_N12345_G2R1_AA00aa_TYP_MOD.typ

Filename will be constructed with the above sequence of info:
The "_" <underscore> character will be used as a field separator

1. YYYY-MM-DDTHHMMSSZ - ISO 8601 format of UTC Date and Time for the collected data start time in file (18 Chars) (no colon ":" character is allowed in the filename)
2. N12345 – Node number assigned to this station's data collection computer - field ends with _
   N – Defines the beginning of the node field
   12345 – 5 digit node number (unique to this data collection computer at this location)
   Nodes 00000 and 99999 have special functions assigned to them and are not for general use

   Modifiers for Node's file data being submitted - G1R1
   A – Alpha character to define the specific data collection platform / Receiver # collecting data
   G – Grape (Low Cost Platform)
   1 – Generation 1 (has 1 receiver)
   2 – Generation 2 (has 4 receivers)
   T – Tangerine (High Performance Platform)
   1 – Generation 1
   2 – Generation 2
   S – Standard / Commercial (Amateur) Hardware (details in Metadata)
   U – User Custom Hardware (details in Metadata)
   R1 – Receiver slice being used for this channel (for multichannel receivers – 1 and up)
   If receiver only has 1 channel (Grape, Standard, User), the R modifier field is omitted

3. AA00aa – GridSquare locator of station (6 Chars) - Autogenerated - field ends with _
4. TYP – 3 (or more) character description of data - field ends with _
   This field is optional and is intended to be used wher it makes sense (i.e. not required)
   - FRQ = frequency data of detected Beacon
   - TMP = Temperature data of station and Magnetometer
   - MAG = Magnetometer data – 3 axis
   - LGT = Lightning strike data
   - DRF = Digital RF file data
   - RFS = RF spectrum database
   - tbd = added as needed

5. MOD – A modifier field to further describe the TYP field. Variable length (0 to ? Chars)
   MOD field starts after TYP field "_" and is terminated by the . for the files extension .csv
   This modifier field is also optional and only used where it makes sense. Grape PWSW uses it to identify the beacon frequency being monitored or other data being collected.
   MOD identifies data type included in the file:
   - FRQ: WWV2p5, WWV5, WWV10, WWV15, WWV20, WWV25, CHU3, CHU7, CHU14

These are initially used by the Grape Gen 1 (and 2) receiver for Beacon identification

- TMP: N/A
- MAG: N/A
- LGT: N/A
- DRF: tbd (?)
- RFS: tbd (?)

6. .typ – Descriptor of file format for data storage
   File Types include (but are not limited to):
   - .csv – comma seperated variable file fomat
   - .txt – text file format
   - .dat – data file format
   - .drf – Digital RF file format
   - .rfs – RF Spectrum file format
   - .wav – Audio format data
   - .raw – raw data format [tbd]
   - .a2d – stored direct A/D readings
   - .tbd - tbd

Data File Naming Examples

Low Cost (Grape) PSWS data file examples:

2020-05-03T000000Z_N00001_G1_EN91fh_FRQ_WWV2p5.csv
05-03-2020 00:00:00Z, Node 00001, Grape Unit 1, EN91fh Grid Square, Freq Data, WWV 2.5 MHz
2020-05-03T000000Z_N00002_G1_EN91fl_FRQ_WWV5.csv
05-03-2020 00:00:00Z, Node 00001, Grape Gen 1, EN91fl Grid Square, Freq Data, WWV 5 MHz
2020-05-03T000000Z_N00003_G1_EN91fh_FRQ_WWV10.csv
05-03-2020 00:00:00Z, Node 00001, Grape Gen 1, EN91fh Grid Square, Freq Data, WWV 10 MHz
2020-05-03T000000Z_N00005_G2R1_AA00aa_FRQ_WWV15.csv
05-03-2020 00:00:00Z, Node 00001, Grape Gen 2, Rcvr 1, AA00aa Grid Square, Freq Data, WWV 15 MHz
2020-05-03T000000Z_N00005_G2R2_AA00aa_FRQ_CHU3.csv
05-03-2020 00:00:00Z, Node 00001, Grape Gen 2, Rcvr 2, AA00aa Grid Square, Freq Data, CHU 3.330 MHz
2020-05-03T000000Z_N00005_G2R3_AA00aa_FRQ_CHU7.csv
05-03-2020 00:00:00Z, Node 00001, Grape Gen 2, Rcvr 3, AA00aa Grid Square, Freq Data, CHU 7.850 MHz
2020-05-03T000000Z_N00005_G2R4_AA00aa_FRQ_CHU14.csv
05-03-2020 00:00:00Z, Node 00005, Grape Gen 2, Rcvr 4, AA00aa Grid Square, Freq Data, CHU 14.670 MHz
2020-05-03T000000Z_N00005_G2_AA00aa_TMP.csv
05-03-2020 00:00:00Z, Node 00005, Grape Gen 2, AA00aa Grid Square, Temp Data
2020-05-03T000000Z_N00005_G2_AA00aa_MAG.csv
05-03-2020 00:00:00Z, Node 00005, Grape Gen 2, AA00aa Grid Square, Magnetometer Data
2020-05-03T000000Z_N00005_G2_AA00aa_LGT.csv
05-03-2020 00:00:00Z, Node 00005, Grape Gen 2, AA00aa Grid Square, Lightning Data

High Performance (Tangerine) PSWS data file examples:

2020-05-03T000000Z_N00010_T1R1_AA00aa_FRQ_WWV2p5.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, Rcvr 1, AA00aa Grid Square, Freq Data, WWV 2.5 MHz
2020-05-03T000000Z_N00010_T1R2_AA00aa_FRQ_WWV5.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, Rcvr 2, AA00aa Grid Square, Freq Data, WWV 5 MHz
2020-05-03T000000Z_N00010_T1R3_AA00aa_FRQ_WWV10.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, Rcvr 3, AA00aa Grid Square, Freq Data, WWV 10 MHz
2020-05-03T000000Z_N00010_T1R4_AA00aa_FRQ_WWV15.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, Rcvr 4, AA00aa Grid Square, Freq Data, WWV 15 MHz
2020-05-03T000000Z_N00010_T1R5_AA00aa_FRQ_WWV20.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, Rcvr 5, AA00aa Grid Square, Freq Data, WWV 20 MHz
2020-05-03T000000Z_N00010_T1R6_AA00aa_FRQ_WWV25.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, Rcvr 6, AA00aa Grid Square, Freq Data, WWV 25 MHz
2020-05-03T000000Z_N00010_T1R7_AA00aa_FRQ_CHU3.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, Rcvr 7, AA00aa Grid Square, Freq Data, CHU 3.33 MHz
2020-05-03T000000Z_N00010_T1R8_AA00aa_FRQ_CHU7.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, Rcvr 8, AA00aa Grid Square, Freq Data, CHU 7.85 MHz
2020-05-03T000000Z_N00011_T1R1_AA00aa_FRQ_CHU14.csv
05-03-2020 00:00:00Z, Node 00011, Tangerine Gen 1, Rcvr 1, AA00aa Grid Square, Freq Data, CHU 14.67 MHz
2020-05-03T000000Z_N00010_T1_AA00aa_TMP.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, AA00aa Grid Square, Temp Data
2020-05-03T000000Z_N00010_T1_AA00aa_MAG.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, AA00aa Grid Square, Magnetometer Data
2020-05-03T000000Z_N00010_T1_AA00aa_LGT.csv
05-03-2020 00:00:00Z, Node 00010, Tangerine Gen 1, AA00aa Grid Square, Lightning Data

PSWS 1st Line Metadata Format

All metadata at the beginning of the data collection file will start with the # <hash> character. The allow the file parser to ignore these line for the actual data collection.

First Row

The first row of the stored *.csv file will contain metadata information about the node collecting the data. It is from these fields that the information presented on the daily data plot will be taken from.

This data is obtained every day from the *.txt files defined below for this station's info. The same file provides the metadata as well so there is no discrepancy in the stored data.

An example of the 1st line [row] of days collection file:

#,2020-05-28THH:MM:SSZ,N00001,EN91fh,41.3219273, -81.5047731,285,Macedonia Ohio,G1,WWV5

Second(+) Rows of Metadata from ~/PSWS/Sinfo/*.txt files

Starting with the 2nd row all data contained about the station will be a human readable text formatted list of the station's info files whose contents (in directory ~/PSWS/Sinfo/ have each line starting with the single character # <hash>

The listing of the station information will contain the following:

- ROW 1 - # used for making parsing easier (defines a metadata line)
- ROW 2 – ISO format DATE/TIME Format: YYYY-MM-DDTHH:MM:SS (19 chars)
- ROW 3 – Node Number for station Format: N00001 (6 chars)
- ROW 4 – Callsign Format: Std Callsign Format using capital letters
- ROW 5 – GridSquare for Station Format: AA00aa (6 chars)
- ROW 6 – Latitude for Station Format: sNN.NNNN (s=sign*, MIN of 4 decimal places)
- ROW 7 – Longitude For Station Format: sNNN.NNNN (s=sign*, MIN of 4 decimal places)
- ROW 8 – Elevation for Station Format: NNNNN (in meters, rounded to nearest meter)
- ROW 9 – City and State for Station Format: City State (separated by 1 space, NO commas)
- ROW 10 – Radio Format: Make Model (separated by 1 space, NO commas)
- ROW 11 – RadioID Format: G2R1 (as defind above) – used in header line
- ROW 12 – Antenna Format: Make Model or brief description (spaces but NO commas)
- ROW 13 – Freq Standard Used Format: type or Make Model (spaces but NO commas)
- ROW 14 – Station Information Fromat: Computer Type Software versions

* - sign indicates:

+ [plus] for Northern Latitude, - [minus] for Southern Latitude, Range is from +90 to -90 degrees
+ [plus] for Eastern Longitude, - [minus] for Western Longitude, Range is from +180 to -180 degrees

Note: ALL stations in the USA are in the Western Longitude Range, so they will all be negative.

Metadata.txt File

Included in the ~/PSWS/Sinfo/ directory is a Metadata.txt file that is intended to be supplemental info concerning this node unique attributes. The is additional information related to the collected data and is not required unless circumstances dictate its use.

It is also valid to have a zero length Metadata,txt file which will result in no Metadata being added to the header info of the days data file contents.

Data Label Header Line (Row) Info

This row of header info defines the columns of stored data for this file. It is defined by the data engine storing the data every second (or specified time period) to the file. It always starts with the ISO time stamp as the first column.

This line is intended to help define the data when a .csv is opened in a program that uses and understands .csv files (Excel, LibreOffice Calc, etc.)

It immediately follows the Metadata field(s) [if any exist]

- COL 1 – ISO Time stamp HH:MM:SS followed by a comma (no need to repeat the date)
- COL 2 – Column 2 Header Info followed by a comma
- COL 3 – Column 3 Header Info followed by a comma
- COL 4 – Column 4 Header Info followed by a comma
- ...
- COL N – Column N added as needed

As an example the different file headers for the Grape platform would look like:

FRQ	Example: UTC,Freq,Vpk
MAG	Example: UTC, X(nT), Y(nT), Z(nT)
TMP	Example: UTC, In Tmp (C), Out Tmp (C)
LGT	Example: UTC, Dist (Mtrs)

After the operational header, the collected data set starts. Each second specified has its own row by itself. Rows do not wrap around to the next row.

Data to be collected and stored every day is a direct measurement from the experiment and NOT information that can be derived from the collected data. This can be done later and it saves space.

For a typical given UTC day, UTC time will go from 00:00:00 to 23:59:59 for a total of 86,400 data points.

Metadata Files and Contents

To assist the user in maintaining this information, the station directory ~/PSWS/Sinfo/ (or usually /home/pi/PSWS/Sinfo/) will contain the following files (and their defined formats):

- **CallSign.txt** – Callsign of the control operator for the station
  Example: N8OBJ
  Example: AD8Y
  Example: AB4EJ

- **NodeNum.txt** – Node number that has been assigned to this station (N12345) (6 chars)
  Example: N00000 – specially designated test node – not for use by any official station
  Example: N00001
  Example: N01234
  Example: N99999 – specially designated cmd node – used to talk to all stations

- **GridSqr.txt** – Maiden-Head Gridsquare Locator for station – 6 Chars in AA00aa format
  This field is auto generated by the Lattitude / Longitude information
  Example: EN91fh
  Example: EN91fl
  Example: EM63fj

- **LatLonElv.txt** – Latitude, Longitude, Elevation of the station as defined above
  A Comma separates the Latitude and Longitude as well as the Longitude and Elevation but NO comma after the Elevation. 4 decimal places minimum. Elevation accuracy is to 1 meter (no decimal places).
  Example: 41.3219273, -81.5047731,285
  Example: 41.3219, -81.5048, 285

- **CityState.txt** – Contains the City and State of the station separated by a space and NO comma
  Example: Macedonia Ohio
  Example: Cleveland Hts Ohio
  Example: Tuscaloosa Alabama

- **FreqStd.txt** – Defines the frequency standard being used to make the measurements
  Options are:
  - XTAL – standard quartz crystal oscillator
  - TCXO – Temperature Compensated Crystal Oscillator
  - OCXO – Oven Compensated Crystal Oscillator
  - Rubidium – Rubidium Frequency Standard (Make Model with spaces but NO commas)
  - GPSDO – GPS Disciplined Oscillator (Make Model with spaces but NO commas)
  - Cesium – Cesium Frequency Standard (Make Model with spaces but NO commas)
  - OTHER – custom Frequency standard (brief description with spaces but NO commas)

  Example: XTAL
  Example: TCXO

Example: OCXO
Example: Rubidium Ball Efratom Division Model MRT
Example: GPSDO Leo Bodnar mini GPS

- **Radio.txt** – Radio Make and Model (spaces but NO comas)
  Example: Grape Gen 1
  Example: Tangerine Gen 1
  Example: ICOM IC-7610
  Example: FlexRadio Flex6500
  Example: Yaesu FT-847
  Example: Kenwood TS-570

- **Antenna.txt** – Make and Model of Antenna used or brief description (with spaces NO commas)
  Example: 135 foot OCF Dipole 30 feet up
  Example: Gap Titan
  Example: Cushcraft R6000

- **Station.txt –** Metadata about station including Computer platoform, OS version, Firmware version(s) of running software
  Example: RasPi4B w/4GB, Raspian Linux Ver 4.19.118-v7l+, FLDigi Ver 4.1.13
  Example: RasPi4B w/4GB, Raspian Linux Ver 4.19.118-v7l+, Grape Software Version 1.02

- **Metadata.txt** – Metadata file for information pertaining to this node. ONLY rule is that All lines start with the char # <hash>. This file can contain as many lines as it wants with any format they choose appropriate as long as the line start with #. This metadat file in included with each days data at the head of the file and is intended to include other

- **Beacon** – This file is autogenerated at the beginning of each UTC day and reflect the detected beacon that the station is currently monitoring. Entries are taken from the freq mod field previously defined and will be one of the following:

  WWV2p5, WWV5, WWV10, WWV15, WWV20, WWV25, CHU3, CHU7, CHU14

File Structure used on the PSWS

This is intended to be running a Linux Kernel (Raspbian) Raspberry Pi for the Grape (Low Cost PSWS) or Linux (Centos) on the Tangerine (High Performance) PSWS.

~ = user home directory for filesystem – which for Raspian User pi would translate to /home/pi

PSWS filenames can include the following characters imbedded into the filenames
- dash
_ underscore

ALL other punctuation and white space characters ARE NOT ALLOWED.
These include, but are not limited to `~!@#$%^&*()=+[]/\{}|:;'"<>,?.

From here:

- ~ is the home directory of the user – such as /home/pi for user pi
- ~/PSWS – Full Base directory path for all PSWS files located on this computer
  for user pi this becomes = /home/pi/PSWS/

Station Subdirectories of ~ /PSWS are as follows:

- /Rawdata - directory where all PSWS day's raw data files [for data processing] are stored
- /Scode - where all code (python and C/C++) is located for data processing
- /Stemp - temp location for data processing
- /Sinfo – Station Information location for metadata about station
- /Sstat - daily cron status location and other status messages
- /Scmd – location for remote command files
- /Splot – local storage for plots of the days data collection results
- /Sdata – storage directory for all processed station data [to be sent back to database]

For the processed Station data, the following subdirectories are for the assigned data being collected. Other subdirectories can be added as needed for other experiments.
- /WWV2p5 - processed data and plots for 2.5MHz WWV data
- /WWV5 - processed data and plots for 5MHz WWV data
- /WWV10 - processed data and plots for 10 MHz WWV data
- /WWV15 - processed data and plots for 15 MHz WWV data
- /WWV20 - processed data and plots for 20 MHz WWV data
- /WWV25 - processed data and plots for 25 MHz WWV data
- /CHU3 - processed data and plots for 3.330 MHz CHU data
- /CHU7 - processed data and plots for 7.850 MHz CHU data
- /CHU14 - processed data and plots for 14.670 MHz CHU data
- /MAG - processed data and plots for Magnetometer data
- /TEMP - processed data and plots for Temperature Data
- /LGTNG - processed data and plots for Lightning Data
- /DRF – processed data for the Digital RF formated Data

- /RFS - processed data for the RF Spectrum formated Data

Directory Structure in Tree Format

The intent of this directory structure is to contain ALL data and programs files related to the running of the PSWS. This makes it more intuitive for the average user who is not a famaliar with the internals of an operating system.

Home directory:
~/ or /home/pi – User Home Directory

MAIN Subdirectory:
/PSWS – Base directory for all PSWS files

PSWS Subdirectories:
/Srawdata – where todays active raw data collection files reside
/Scode – where the python scripts and data collection programs reside
/Stemp – temp working directory for file manipulation
/Sinfo – contains all the metadata files realted to this node
/Sstat – contains status info on task related to this station
/Scmd – contains the commands received from the central data collection node
/Splot – local storage for plots of the days data collection results
/Sdata – contains the processed files that will be sent back to the central collection node

/Sdata Subdirectories:
/WWV2p5 – processed files for the WWV 2.5 MHz Beacon
/WWV5 – processed files for the WWV 5 MHz Beacon
/WWV10 – processed files for the WWV 10 MHz Beacon
/WWV15 – processed files for the WWV 15 MHz Beacon
/WWV20 – processed files for the WWV 20 MHz Beacon
/WWV25 – processed files for the WWV 25 MHz Beacon
/CHU3 – processed files for the CHU 3.33 MHz Beacon
/CHU7 – processed files for the CHU 7.85 MHz Beacon
/CHU14 – processed files for the CHU 14.67 MHz Beacon
/MAG – processed files for the 3 axis magnetometer
/TEMP – processed files for the Temperature Sensors
/LGTNG – processed files for the Lightning Sensor
/DRF – processed files for the Digital RF files
/RFS – processed files for the RF Spectrum Files

A Day in the Life of a Grape Gen 1 PSWS

This narrative is a brief description of how data is collected and processed on a daily basis for the Grape Gen 1 PSWS. It is intended to be an overview of how things operate and where files are placed for normal operation. THIS IS NOT intended to cover everything that is happening, but it gives a good overview of what was intended to be implemented and how it was intended to work. I wrote this narrative as I was coding up the running programs so the details were fresh and accurate. Don't ask me to rewirite this in 4 weeks...

This process will change as the project matures and evolves. A similar narrative will be included for the Tangerine PSWS at a later date(?).

Soooooooo, the operating system time hits 00:00:00 UTC...

This kicks off a new data collection file by FLDigi ver 4.1.13 (customized by N8OBJ) [or a custom data collection program – coming soon to a Grape Gen2 PSWS near you!] running on this node to start a new days data collection file. This data file is stored in ~/PSWS/Srawdata/ as file analysisYYMMDD.csv where YYMMDD is today's new UTC date. The program then determines the beacon being monitored and checks and updates [if needed] the ~/PSWS/Sinfo/Beacon file with the beacon identifier string (WWV5 for example). By only upgrading the Beacon file when the frequency changes, you can determin how long you have been monitoring this particular beacon by looking at the Beacon's file creation date(!). The program also reads system related information (Node Number, Gridsquare, Latitude / Longitude / Elevation, City State, RadioID, Beacon1) from this same /Sinfo directory. This is done to create the first line header info to aid in the plotting of this data file the following day.

Data collection continues in this file for the entire UTC day. If the data collection process is interrupted or stopped, the data is saved (even on a system crash with some 'issues' to be addressed later in post-processing) in the filesystem. And I've never seen it loose data on a system crash. When the process restarts the new data gets appended to this same analysisYYMMDD.csv file where the data collection was interrupted as long as the UTC date is still the same. This allows for system down time, maintaince, system crashes, etc.

The beacon frequency data is created and stored on a one second epoch time interval to the .csv file by the data collection program. This gives 86,400 data entries per day in this file. On a Grape Gen 2, there will be 3 other receiver channels collecting and storing data. In addition, there will be a 3 axis magnetometer and 2 temperature sensors (inside and outside) having their data collected as well (into appropriate different files). But for now, Grape Gen 1 just has the one receive frequency channel.

During the course of the day, either at a pre-arranged time(s) or time interval(s) the station pings the central data collection node for any commands that may have been queued for it. If there are any, they will be received and placed in directory ~/PSWS/Scmd with a file name structure as

cmd-N12345-YYYY-MM-DDT00:00:00Z
where:

cmd – indicates this is a command file from the central server
N12345 – this node's number

YYYY-MM-DDT00:00:00Z – date and time the command was issued

The system response will be to store the command file in the ~/PSWS/Scmd/ directory and will execute its contents (at a time determined by local control of the OS) ONLY if the node number matches the one assigned to this node. Upon completion of execution, the node will rename this file by adding a ".done" to the filename in the same /Scmd directory. This technique creates a log of files of the commands it received and executed in this directory for future reference and debugging. It can also send an acknowledgement to the central control node that the command was completed. If a status handshake or file is to be sent, the status of this transaction will be saved in the ~/PSWS/Sstat/ (or ~/PSWS/Scmd/) directory (tbd). For command files, ths would be the same filename only that the files extension will be ".stat" instead of ".done". A new time stamp is not necessary as it will be contained in the file system info for the stored file.

A special case of "COMMAND ALL NODES" will be assigned to node N99999. This would be used for a universal braoadcast command for the whole network.

Data collection continues throughout the day until the system clock hits 00:00:00 UTC again.

Then the whole process for next UTC day starts all over again.

But also at 00:00:00 UTC a crontab job for the user pi kicks off a python program to check the data file for corruption and to rename and move it to the correct directory for submission to the central node server. The script also removes the Freq Err and dB(Vpk) data colums from the file to be submitted to the server as well (these numbers can be derived from the data already there and are therefore redundant and not necessary).

This script is implemented on the RasPi for user pi by the following command:
$ sudo crontab -u pi -e

The entry put into this user pi crontab file looks like:

00 00 * * * "python3 /home/pi/PSWS/Scode/WWVfilechk.py >/home/pi/PSWS//Sstat/wwvcronstat 2>&1; export DISPLAY=:0 && python3 /home/pi/PSWS/Scode/WWV_plt2.py >/home/pi/PSWS//Sstat/wwvcronplot 2>&1"

- The export DISPLAY=:0 is needed to open the plotting window for the python3 run a plot
- The output of the script (as it runs) is redirected to file wwvcronstat in the proper directory
- Standard Linux output (1) and error status output (2) channels are both sent to this file with 2>&1

To view the crontab job(s) for user pi, use the command:
$ sudo crontab -u pi -l

To view the crontab job(s) for user root, use the command:
$ sudo crontab -u root -l

We are currently using a modified version of FLDigi Ver 4.1.13 for now. Its purpose is to collect the 1 sec epochs of data, analyse it to determine the beacon center frequency and save the UTC time, freq,

freq error, Vpk and dBV(Vpk) per sample. This is a temporary solution as the final version will also be a compiled C coded program to save the same data for daily processing. This was done to get the system up and running for faster prototyping reasons. I [N8OBJ] have been saving data on node N00001 since 2019-05-24 using this method (I started with FLDigi Ver 4.1.09).

The header and first few samples from a saved day's file from FLDigi now looks like the following:

```
#,2020-05-28T00:00:00Z,N00001,EN91fh,41.3219273, -81.5047731, 285,Macedonia Ohio,G1,WWV5
# metadata starts here (for as many lines as needed) – all lines start with #
UTC,Freq,Freq Err,Vpk,dBV(Vpk)
00:00:00,   4999999.732, -0.268, 0.009360, -40.57
00:00:01,   4999999.638, -0.362, 0.012044, -38.38
00:00:02,   4999999.821, -0.179, 0.011693, -38.64
00:00:03,   4999999.872, -0.128, 0.012090, -38.35
00:00:04,   4999999.902, -0.098, 0.011122, -39.08
00:00:05,   4999999.877, -0.123, 0.010954, -39.21
00:00:06,   4999999.851, -0.149, 0.012105, -38.34
00:00:07,   4999999.823, -0.177, 0.011360, -38.89
00:00:08,   4999999.843, -0.157, 0.011423, -38.84
00:00:09,   4999999.848, -0.152, 0.010891, -39.26
00:00:10,   4999999.707, -0.293, 0.012743, -37.89
```

This file format of saved data is what I submitted to the author of FLDigi (David – W1HKJ) which he included in release 4.1.12 and I am using this raw data format for compatability of future data colletion participants by a station running FLDigi and their own radio. The basic issues in FLDigi I fixed with this submission to W1HKJ were:

1 – When you restart the analysis mode, you don't loose all previously stored data
2 – The UTC date is tacked onto the end of the analysis filename for long term data collection
3 – At 00:00:00 UTC a new file is created and the previous day's data is saved and kept seperate
4 – Relative timestamping was removed and we just used the absolute UTC timestamp
5 – The file is put into a directory that is not hidden (in Linux, the - /.fldigi directory is hidden, but not sure he kept this new feature I added)

The file processsing at the end of the UTC day first just copies all of the metadata lines (lines that start #) to the new file. It then picks out the UTC, Frequency and Vpk fields from the above saved days data file and saves them until all data is transferred. This new file is then renamed to a filename with the format defined above. Finally, it is moved to the ~/PSWS/Sdata/Beacon directory where Beacon is the decoded beacon type (i.e. WWV5).

This file renaming will have the following filename:

2020-05-28T000000Z_N00001_G1_EN91fh_FRQ_WWV5.csv

This file will then be moved to directory: ~/PSWS/Sdata/WWV5/

The processed file structure inside the file looks like the following:

```
#,2020-05-28T00:00:00Z,N00001,EN91fh,41.3219273, -81.5047731, 285,Macedonia Ohio,G1,WWV5
# metadata starts here (for as many lines as needed) – all lines start with #
UTC,Freq,Vpk
00:00:00,   4999999.732, 0.009360
00:00:01,   4999999.638, 0.012044
00:00:02,   4999999.821, 0.011693
00:00:03,   4999999.872, 0.012090
00:00:04,   4999999.902, 0.011122
00:00:05,   4999999.877, 0.010954
00:00:06,   4999999.851, 0.012105
00:00:07,   4999999.823, 0.011360
00:00:08,   4999999.843, 0.011423
00:00:09,   4999999.848, 0.010891
00:00:10,   4999999.707, 0.012743
```

In addition, the previous day's data needs to be checked for known types of corruption (*1*), named correctly and then stored in the correct directory for uploading to the central server.

This is accomlished by taking the previous UTC days analysis file and scanning it for the null character (hex 00). If none are found, then the file can proceed in the file renaming and moving operations.

If a null characer was detected, a temporaty file with the same name is created in the ~/PSWS/Stemp/ directory. The total file length in bytes is determined and then the original file in ~/PSWS/Srawdata/ is parsed in binary mode looking for the hex 00 null character. If the binary character read is found to be non-zero, it is written to the new image in the /Stemp/ directory where null chars are just skipped and not saved. This process continues until all the bytes in the file have been checked. When finished, the file is re-parsed for time stamps and are checked and resequenced (if necessary). At this point the new file is then processed as normal. The newly named file then gets moved (and plotted) to the appropriate directory for transfer to the central node. Status of this whole operation is saved in the ~/PSWS/Sstat/ directory called wwvcronstat [since it was kicked off by a crontab job for user pi] on a daily basis [overwriting the previous day's status file as the crontab filename for the status filename is hard coded in crontab].

Renaming of the file consistes of embedding the date / time, Node number, Radio ID, Gridsquare, Beacon name, and correct file extension for the saved data. It then is placed into ~/PSWS/Sdata/ directory under the correct sudirectory for the collected data (such as /WWV5 for the 5 MHz WWV beacon data).

A graph of the data just processed is created and stored in the ~/PSWS/Splot/ directory and a process is also invoked to plot the graph and display it on the system desktop using the gpicview utility program. Since all of the plots are stored in the same directory the user can use the fwd / rev arrows (in the lower menu bar of the gpicview program window) to look at other days data in the same window.

This processing will also happen for any of the other data collected during the day for any other data collection tasks (magnetometer, temperature, etc). When completed, a request is generated and sent to the central data collection node. Also, status of the operation for this day is saved in the ~/PSWS/Sstat

(*1*) It has been observed that when a RaspBerry Pi looses power without being shut down correctly the data files that were being written to in the filesystem keep all of their data, but the last file block contents for the last file sector on the disk are padded with null characters (hex 00). Then when the system restarts and data collection also restarts, this null block is kept and a data is saved in a new sector following the end of the existing file. This then captureds this block of null characters in the middle of the file. The time stamp for the system is also randomly set (usually backwards) in time until then clock syncs itself up and jumps to the correct time. This block of time shifted data in the file [represents a tear in the space-time continium] and needs to be fixed, but the presence of null chars in the file will cause the python parser to crash in .csv read mode.

The fix for this is a 2 step process: First, all of the null characters are simply removed from the file. Second, from where the time stamp jumps the first time until the 2nd time jump, the system has to parse forward from this point until the 2nd time jump is found, then backstep the time stamps in the file from where this 2nd time jump happened by walking backwards 1 second at a time from this point in the file. If this is a quick system powerfail / reboot, there usually ends up being about a 10-15 second gap in time for the actual system reboot and doesn't really impact the data plotting very much.

You can see that the data collection's last valid entry was at 19:49:36 when power went down unexpectidly. The last buffer of data was written to disk but the end-of-file pointer was not updated in the file allocation table.

When the system reboots, the OS assumes that the whole block is valid data and hence captures the null chars inside the file. Writing starts back up at the next assigned disk block on the disk and the time stamp gets moved back to 19:17:20. This continues until 19:18:21 (61 seconds later).

Data continues to be collected normally until the system clock finally synchronizes itself at 19:51:28 at which point the time jumps forward to the correct time. We now hae null chars imbedded in the file with 61 seconds of time shifted data.

This is the fingerprint of a system power failed shutdown without sequencing down properly.

The attached hex data dump listing of a corrupted file shows what is going on when the system shuts down improperly.

Replace All | Replace | Replace & Find | Previous | Next

```
5, -63.42 19:49:15,   4999999.991, -0.009, 0.000696, -63.15 19:49:16,   4999999.979, -0.021, 0.00070
1, -63.09 19:49:17,   4999999.745, -0.255, 0.000680, -63.34 19:49:18,   4999999.605, -0.395, 0.00085
4, -61.37 19:49:19,   4999999.762, -0.238, 0.000868, -61.23 19:49:20,   4999999.750, -0.250, 0.00089
7, -60.94 19:49:21,   5000000.040,  0.040, 0.000950, -60.44 19:49:22,   5000000.067,  0.067, 0.00077
0, -62.27 19:49:23,   5000000.018,  0.018, 0.000775, -62.21 19:49:24,   5000000.021,  0.021, 0.00080
1, -61.93 19:49:25,   4999999.965, -0.035, 0.000795, -62.00 19:49:26,   4999999.971, -0.029, 0.00078
8, -62.07 19:49:27,   4999999.960, -0.040, 0.000769, -62.29 19:49:28,   4999999.990, -0.010, 0.00076
8, -62.30 19:49:30,   5000000.014,  0.014, 0.000788, -62.07 19:49:31,   5000000.022,  0.022, 0.00079
2, -62.02 19:49:32,   5000000.066,  0.066, 0.000820, -61.72 19:49:33,   4999999.999, -0.001, 0.00082
7, -61.65 19:49:34,   5000000.028,  0.028, 0.000830, -61.62 19:49:35,   5000000.001,  0.001, 0.00081
3, -61.80 19:49:36,   5000000.047,  0.047, 0.000825, -61.67
                                                                        19:17:20,   4999999.921, -0.079, 0.00038
6, -68.27 19:17:21,   4999999.918, -0.082, 0.000607, -64.33 19:17:22,   4999999.875, -0.125, 0.00081
4, -61.79 19:17:23,   5000000.045,  0.045, 0.000799, -61.95 19:17:24,   5000000.017,  0.017, 0.00077
8, -62.18 19:17:25,   5000000.028,  0.028, 0.000744, -62.57 19:17:26,   5000000.029,  0.029, 0.00074
5, -62.56 19:17:27,   4999999.996, -0.004, 0.000750, -62.50 19:17:28,   4999999.963, -0.037, 0.00071
9, -62.87 19:17:29,   4999999.964, -0.036, 0.000701, -63.09 19:17:30,   5000000.056,  0.056, 0.00068
0, -63.35 19:17:31,   4999999.987, -0.013, 0.000662, -63.58 19:17:32,   5000000.008,  0.008, 0.00066
4, -63.55 19:17:33,   5000000.017,  0.017, 0.000691, -63.21 19:17:34,   4999999.996, -0.004, 0.00071
7, -62.89 19:17:35,   4999999.980, -0.020, 0.000751, -62.48 19:17:36,   5000000.039,  0.039, 0.00076
1, -62.37 19:17:37,   4999999.978, -0.022, 0.000734, -62.68 19:17:38,   5000000.023,  0.023, 0.00071
2, -62.96 19:17:39,   5000000.050,  0.050, 0.000672, -63.45 19:17:40,   5000000.014,  0.014, 0.00063
8, -63.90 19:17:41,   4999999.932, -0.068, 0.000570, -64.88 19:17:42,   4999999.744, -0.256, 0.00049
2, -66.16 19:17:43,   4999999.730, -0.270, 0.000474, -66.48 19:17:44,   4999999.716, -0.284, 0.00047
7, -66.43 19:17:45,   4999999.715, -0.285, 0.000533, -65.46 19:17:46,   4999999.962, -0.038, 0.00059
5, -64.51 19:17:47,   5000000.015,  0.015, 0.000650, -63.74 19:17:48,   4999999.955, -0.045, 0.00070
1, -63.09 19:17:49,   5000000.071,  0.071, 0.000733, -62.70 19:17:50,   5000000.059,  0.059, 0.00073
0, -62.74 19:17:51,   5000000.053,  0.053, 0.000729, -62.75 19:17:53,   5000000.009,  0.009, 0.00066
4, -63.56 19:17:54,   4999999.985, -0.015, 0.000668, -63.50 19:17:55,   4999999.992, -0.008, 0.00071
3, -62.94 19:17:56,   5000000.009,  0.009, 0.000725, -62.79 19:17:57,   5000000.003,  0.003, 0.00082
2, -61.70 19:17:58,   4999999.967, -0.033, 0.000807, -61.86 19:17:59,   4999999.979, -0.021, 0.00077
9, -62.17 19:18:00,   4999999.985, -0.015, 0.000798, -61.95 19:18:01,   4999999.996, -0.004, 0.00080
2, -61.92 19:18:02,   4999999.956, -0.044, 0.000859, -61.32 19:18:03,   4999999.992, -0.008, 0.00084
8, -61.43 19:18:04,   5000000.015,  0.015, 0.000810, -61.83 19:18:05,   5000000.028,  0.028, 0.00078
4, -62.12 19:18:06,   5000000.017,  0.017, 0.000705, -63.04 19:18:07,   5000000.025,  0.025, 0.00072
9, -62.75 19:18:08,   5000000.014,  0.014, 0.000731, -62.72 19:18:09,   5000000.020,  0.020, 0.00071
4, -62.92 19:18:10,   5000000.004,  0.004, 0.000764, -62.33 19:18:11,   5000000.017,  0.017, 0.00079
1, -62.03 19:18:12,   4999999.982, -0.018, 0.000858, -61.33 19:18:13,   4999999.970, -0.030, 0.00088
0, -61.11 19:18:14,   4999999.960, -0.040, 0.000877, -61.14 19:18:15,   4999999.981, -0.019, 0.00083
5, -61.57 19:18:16,   5000000.019,  0.019, 0.000810, -61.83 19:18:17,   4999999.684, -0.316, 0.00133
5, -57.49 19:18:18,   4999999.991, -0.009, 0.001369, -57.27 19:18:19,   4999999.983, -0.017, 0.00139
7, -57.09 19:18:20,   4999999.993, -0.007, 0.001454, -56.75 19:18:21,   5000000.263,  0.263, 0.00100
4, -59.96 19:51:28,   4999999.975, -0.025, 0.001095, -59.21 19:51:29,   4999999.973, -0.027, 0.00114
6, -58.82 19:51:30,   4999999.979, -0.021, 0.001133, -58.91 19:51:31,   4999999.972, -0.028, 0.00115
1, -58.78 19:51:32,   5000000.014,  0.014, 0.001106, -59.12 19:51:33,   5000000.020,  0.020, 0.00106
8, -59.43 19:51:34,   5000000.036,  0.036, 0.001054, -59.54 19:51:35,   5000000.053,  0.053, 0.00096
6, -60.30 19:51:36,   5000000.028,  0.028, 0.000884, -61.08 19:51:37,   5000000.049,  0.049, 0.00083
6, -61.55 19:51:38,   5000000.090,  0.090, 0.000797, -61.98 19:51:39,   5000000.090,  0.090, 0.00080
0, -61.94 19:51:40,   5000000.088,  0.088, 0.000815, -61.78 19:51:41,   5000000.108,  0.108, 0.00077
1, -62.26 19:51:42,   4999999.987, -0.013, 0.000695, -63.16 19:51:43,   4999999.741, -0.259, 0.00058
6, -64.64 19:51:44,   4999999.463, -0.537, 0.000493, -66.14 19:51:45,   4999999.481, -0.519, 0.00043
8, -67.16 19:51:46,   4999999.495, -0.505, 0.000431, -67.32 19:51:48,   4999999.707, -0.293, 0.00048
9, -66.22 19:51:49,   4999999.982, -0.018, 0.000537, -65.39 19:51:50,   4999999.973, -0.027, 0.00059
0, -64.58 19:51:51,   4999999.975, -0.025, 0.000576, -64.79 19:51:52,   5000000.000, -0.000, 0.00055
3, -65.14 19:51:53,   5000000.011,  0.011, 0.000521, -65.66 19:51:54,   4999999.983, -0.017, 0.00051
8, -65.71 19:51:55,   4999999.998, -0.002, 0.000553, -65.14 19:51:56,   5000000.016,  0.016, 0.00063
5, -63.95 19:51:57,   5000000.043,  0.043, 0.000718, -62.88 19:51:58,   4999999.979, -0.021, 0.00080
8, -61.85 19:51:59,   4999999.966, -0.034, 0.000860, -61.31 19:52:00,   5000000.025,  0.025, 0.00089
```